Measurability of a Map

Definition

Let $(X, \mathcal{A})$ and $(Y, \mathcal{B})$ be measurable spaces. A map

$$T : X \to Y$$

is measurable if

$$T^{-1}(B) \in \mathcal{A}$$

for every $B \in \mathcal{B}$.

Special Case

If $X = \mathbb{R}^m$ and $Y = \mathbb{R}^d$ are equipped with their Borel σ-algebras, then $T : \mathbb{R}^m \to \mathbb{R}^d$ is measurable if

$$T^{-1}(B)$$

is a Borel subset of $\mathbb{R}^m$ for every Borel subset $B \subseteq \mathbb{R}^d$.